

Supplementary Table S4. Model selection summary

Comparison	Hypothesis	Result	Interpretation
Model 1 vs Null	Do genetic factors alone explain variation?	No (p = 0.122)	Genetics alone insufficient
Model 2 vs Model 1	Do sex and infection explain variation?	Yes - Infection matters (p < 0.001)	Infection is primary driver
Model 3 vs Model 2	Do interactions improve the model?	Yes - Sex × HI interaction (p = 0.011)	Males specifically affected by ancestry